

SRI DEVI ARTS AND SCIENCE COLLEGE

DEPARTMENT OF COMPUTER SCIENCE

EDUGENI

Google Gemini Powered Learning Assistant

SWTID : 2026-8737

TEAM MEMBERS

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SUBMITTED BY DEPARTMENT OF COMPUTER SCIENCE

1. PROJECT OVERVIEW

EduGeni is a Google Gemini powered learning assistant designed to support students in understanding academic topics through conversational Generative AI. It can accept questions, learning requirements, and contextual information and generate explanations in a structured and student-friendly manner.

The system aims to make learning more interactive by supporting explanations, summaries, examples, question generation, revision support, and personalized learning guidance.

2. PROBLEM STATEMENT

Students often spend considerable time searching for explanations, examples, revision notes, and practice questions from multiple sources. Information may be difficult to understand when it is presented at an unsuitable level or without enough context.

EduGeni addresses this challenge by providing an AI-based learning assistant that interprets a student's request and generates a suitable educational response. The assistant is intended to support learning while encouraging students to verify important academic information.

3. PROJECT OBJECTIVES

- Develop a Generative AI based learning assistant using Google Gemini.
- Provide clear and structured explanations for academic questions.
- Support summaries, examples, revision notes, and practice questions.
- Allow students to interact using natural-language questions.
- Adapt responses according to the learner's stated requirement or level.
- Provide a simple workflow for learning and revision.

4. SCOPE OF THE PROJECT

EduGeni can be used as a supplementary learning tool for students. It can help explain concepts, summarize study material, create practice questions, provide examples, and assist with revision planning.

The assistant is not intended to replace teachers, textbooks, laboratory work, or official academic resources. Students should verify important facts and follow their institution's academic requirements.

5. SYSTEM INPUTS

The system can accept natural-language questions and learning requests. Example inputs include a topic name, a question, a request for a simple explanation, a request for examples, or a request to create quiz questions.

Students can also specify preferences such as beginner-level explanation, short notes, step-by-step explanation, or exam-oriented revision.

6. GOOGLE GEMINI & GENERATIVE AI

EduGeni uses the capabilities of Google Gemini to interpret user prompts and generate educational content. Generative AI can identify the intent of a question, organize relevant concepts, and produce a natural-language response.

Prompt design is important because clear instructions about topic, level, response length, examples, and output format can improve the usefulness of the generated answer.

7. LEARNING FEATURES

EduGeni can support multiple learning activities. It can explain difficult concepts in simpler language, summarize longer topics, provide examples, create revision points, and generate practice questions.

A student may ask for a topic to be explained step by step or request a short exam revision format.

8. PERSONALIZED LEARNING

Personalization allows the student to state how they want to learn. For example, a user can request a beginner-friendly explanation, a detailed technical explanation, bullet-point notes, or question-and-answer practice.

The assistant can use the information provided in the current interaction to tailor the response.

9. QUIZ & PRACTICE SUPPORT

EduGeni can generate practice questions from a specified topic. Depending on the request, it can produce multiple-choice questions, short-answer questions, or revision questions.

Practice content can help students identify areas that require more study. Answers and explanations should be reviewed when accuracy is important.

10. SYSTEM WORKFLOW

1. Student opens the EduGeni learning assistant. 2. Student enters a topic or academic question. 3. Student optionally specifies learning level and response format. 4. The request is sent to the Generative AI model. 5. Gemini interprets the prompt and generates a response. 6. The response is displayed in an organized format. 7. The student reviews, asks follow-up questions, and continues learning.

11. TOOLS AND TECHNOLOGIES

The project is based on Generative AI with Google Gemini. A suitable application interface can be used to collect student questions and display responses. GitHub can be used to organize source code and project files.

Presentation software is used for the project PPT, while documentation tools are used for the final report.

12. TEAM RESPONSIBILITIES

S. Sneha (22250900) is the team leader and is responsible for coordinating the project and supporting implementation activities.

D. Kalaiyarasi (22250113) is responsible for the document, including project introduction, objectives, methodology, screenshots, results, and conclusion.

T. Sneha (22250901) is responsible for PPT preparation and presentation structure.

All team members should review the final outputs before submission.

13. TESTING AND VALIDATION

Testing should include different academic questions, explanation levels, summary requests, and practice-question requests. The team should check whether the assistant follows the user's instructions and presents responses clearly.

Generated answers should be reviewed for factual accuracy and relevance.

14. RESULTS AND FUTURE ENHANCEMENT

The expected result is an interactive learning assistant that helps students obtain explanations and study support through natural-language interaction. The project demonstrates a practical educational application of Generative AI.

Future enhancements may include multilingual learning, document or image-based question answering, progress tracking, personalized study plans, voice interaction, subject-specific modes, and stronger source verification.

15. CONCLUSION

EduGeni demonstrates how Google Gemini and Generative AI can be used to create a student-focused learning assistant. By accepting natural-language questions and learning preferences, the system can provide explanations, summaries, examples, revision support, and practice questions.

The team has divided responsibilities across leadership, documentation, and PPT preparation. The final project should be reviewed collectively before submission under SWTID 2026-8737.